ZCAS UNIVERSITY

BACHELOR OF SCIENCE IN COMPUTING WITH BUSINESS

CIT 3241 BUSINESS SYSTEM ANALYSIS AND DESIGN

FINAL EXAMINATIONS

FRIDAY, 29TH MAY, 2026

TIME: 09:00 – 12:00 HOURS

TIME ALLOWED: THREE (03) HOURS

INSTRUCTIONS TO CANDIDATES:

1. **SECTION A:** This section is **compulsory** with **ONE (01)** Question and carries **40 marks.**
2. **SECTION B:** Attempt **any THREE (03)** questions from this section. Each question carries **20 marks**.
3. This examination paper carries a total of **100 marks.**
4. Time allowed: **THREE (03) Hours plus FIVE (05) minutes to read through the paper.**
5. **DO NOT** turn this page **until** you are told to do so.

"From Chaos to Agile: Systems Transformation at ZamTech Solutions"

ZamTech Solutions, a mid-sized Zambian software company, had been struggling with frequent project failures. Over 50% of their IT projects were delivered late and significantly over budget, largely because their Systems Analysts failed to properly gather and document requirements before development began. The company had relied heavily on the Waterfall development methodology for years, resulting in long delivery timelines and products that rarely matched what clients actually needed.

Following a critical failed project for a government client — where the delivered system met only 40% of the stated requirements — the CEO commissioned a full review of the company's systems development approach. The review team identified key failures: analysts had skipped formal requirements determination, used only unstructured interviews without follow-up, and had not engaged stakeholders consistently throughout the project. There was no use of techniques such as prototyping, JAD sessions, or questionnaires.

The management decided to transition to Agile development, beginning with Scrum. A Product Owner was appointed to maintain the product backlog, and a Scrum Master was assigned to facilitate sprint ceremonies. The development team was reorganized into cross-functional units tasked with delivering working software in two-week iterations. Additionally, a new requirements elicitation framework was adopted, combining structured interviews, observation, document analysis, and prototyping.

During the first sprint, the Scrum Master facilitated a sprint planning meeting where user stories were prioritized using business value. Daily standups were held to track progress and surface blockers. At the end of the sprint, a demo was held with the client, and a retrospective was conducted to identify process improvements for the next iteration.

Despite this progress, some senior developers resisted the change, citing concerns about Extreme Programming (XP) practices such as pair programming and test-driven development. The team had also not yet established a clear Definition of Done, which led to inconsistencies in what was considered completed work.

Required:

(a) Using the case of ZamTech Solutions, explain why requirements determination is considered the most critical phase of the Software Development Life Cycle (SDLC). Support your answer with specific evidence from the case. *(10 marks)*

(b) The ZamTech review team identified that analysts had only used unstructured interviews. Compare and contrast structured, semi-structured, and unstructured interviews as requirements elicitation techniques, and justify which combination would have been most appropriate for ZamTech's government client project. *(10 marks)*

(c) ZamTech adopted Scrum as its Agile framework. Describe the roles of the Product Owner, Scrum Master, and Team Members, and explain how each role contributed to the success of the first sprint described in the case study. (10 marks)

(d) Analyze the resistance to Extreme Programming (XP) practices at ZamTech. Discuss the principles of pair programming and test-driven development, and argue for or against their adoption at ZamTech, with reference to at least two other XP practices that could benefit the organization.

(10 marks)

(Total : 40 Marks)

SECTION B

CHOOSE ANY THREE (03) QUESTIONS FROM THIS SECTION

QUESTION TWO

(a) Describe the Systems Development Life Cycle (SDLC) and explain the key activities carried out during the Planning and Analysis phases. *(5 marks)*

(b) Compare the Waterfall development methodology with Agile methodology, highlighting one advantage and one disadvantage of each. *(5 marks)*

(c) A retail company in Lusaka wants to build a new inventory management system but has very unclear user requirements and a short delivery timeline. Using the criteria for selecting a development methodology, recommend the most suitable methodology for this company and justify your answer. *(5 marks)*

(d) Explain the following Agile Development principles and state why each is important in software development:

i) Working software over comprehensive documentation
ii) Individuals and interactions over processes and tools

(5 marks)

(Total: 20 marks)

QUESTION THREE

(a) Discuss the importance of Test-Driven Development (TDD) in Extreme Programming (XP) and explain how it improves software quality. (5 Marks).

(b) Explain the role of Pair Programming in Extreme Programming (XP) and describe any three benefits of using this practice in software development. (5 Marks)

(c) Explain the role of prototyping in requirements elicitation and discuss how low-fidelity and high fidelity prototypes help stakeholders refine system requirements. (5 Marks)

(d) Compare traditional and modern methods of requirements elicitation, highlighting two advantages and two limitations of each approach. (5 Marks)

Total: 20 Marks)

QUESTION FOUR

Many organizations continue to struggle with requirements elicitation failures. Research shows that up to 60% of project failures stem from poor requirements gathering, leading to costly rework, scope creep, and user dissatisfaction. Effective elicitation requires the right combination of techniques, skilled analysts, and active stakeholder participation.

(a) Explain the difference between Functional Requirements and Non-Functional Requirements. Provide two examples of each in the context of a student management system. *(5 marks)*

(b) Compare the Joint Application Development (JAD) technique with Focus Groups as requirements elicitation methods. Identify one advantage and one disadvantage of each. *(5 marks)*

(c) A systems analyst is tasked with gathering requirements for a new e-commerce platform. Explain how they could use **Document Analysis** as an elicitation technique. What are the main advantages and limitations of this approach? *(5 marks)*

(d) Describe the Hawthorne Effect and explain how it can influence the outcomes of Observation as a requirements elicitation technique. How can an analyst minimize its impact?

(5 marks)

(Total: 20 Marks)

QUESTION FIVE

(a) Explain the concept of Extreme Programming (XP) and describe the problems with both the Waterfall Model and the Spiral Model that led to its development. *(7 marks)*

(b) Describe any FOUR XP practices and explain how they work together synergistically to improve software quality. Use the concept of synergy as described by Kent Beck which states that "the sum is bigger than the parts. *(6 marks)*

(c) Explain the XP practice of 'On-Site Customer' and discuss its advantages for reducing business misunderstandings. How does this relate to the modern Agile principle of customer collaboration? (7marks)

(Total : 20 Marks)

END OF EXAMINATION PAPER